

special.R

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```
### Special Assignment ###
```

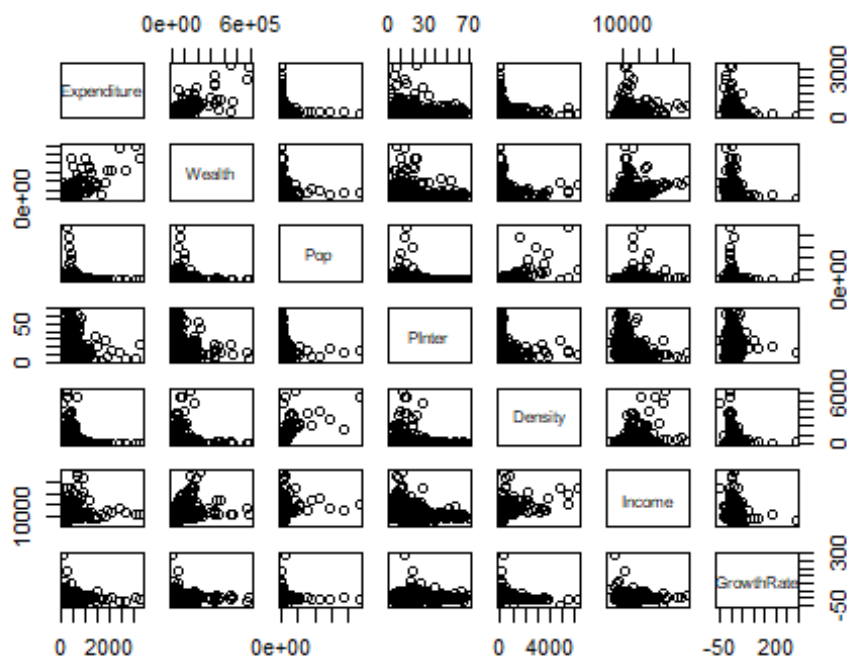
```
#Importing the text file into R
```

```
library(alr4)
```

```
m1 <- read.table("nym.txt",header = TRUE)
```

```
# Regression model for the variables in the data.
```

```
pairs(Expenditure~Wealth+Pop+PInter+Density+Income+GrowthRate,data = m1)
```



```
# as we know that there is no linear relation from OBS,country and state,  
with
```

```
# expenditures so i removed the variables and plotted.By analyzing the plot i  
see some
```

```
# relation with wealth,income.so, Lets summarize the results
```

```
#fitting the regression model by expenditure as response
```

```
m2<-lm(Expenditure~Wealth+Pop+PInter+Density+Income+GrowthRate,data = m1)
summary(m2)
```

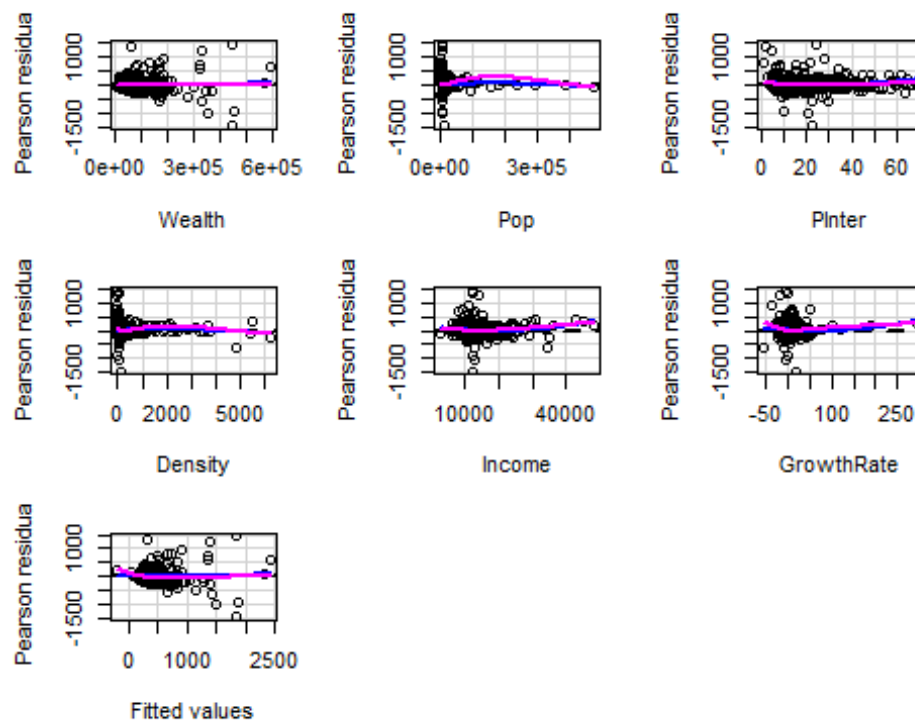
```
##
## Call:
## lm(formula = Expenditure ~ Wealth + Pop + PInter + Density +
##      Income + GrowthRate, data = m1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1449.66   -67.11   -13.12    46.30   1437.89
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.396e+02  2.549e+01   9.399  < 2e-16 ***
## Wealth       3.879e-03  1.083e-04  35.800  < 2e-16 ***
## Pop          3.057e-05  2.881e-04   0.106   0.916
## PInter       1.780e-01  5.820e-01   0.306   0.760
## Density      7.917e-03  1.576e-02   0.502   0.616
## Income      -1.107e-02  1.660e-03  -6.668 4.49e-11 ***
## GrowthRate  -1.366e+00  3.274e-01  -4.173 3.29e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 170.7 on 907 degrees of freedom
## (2 observations deleted due to missingness)
## Multiple R-squared:  0.6021, Adjusted R-squared:  0.5995
## F-statistic: 228.8 on 6 and 907 DF,  p-value: < 2.2e-16
```

```
qt(1-0.05/2,907)
```

```
## [1] 1.962583
```

*#According to the summary,
the given t-value for the wealth,income and growthRate(36,6.6,4.2) is more
than the significant level of quantile
$\alpha=0.05$ which is 1.96.so these regressors are statistically significant also
the p-values of these regressors are
nearly zero.The given data provides evidence against the null hypothesis.
so, we reject the null hypothesis
which states that there is some effect between expenditure and predictors-
wealth,income and growthRate.*

```
residualPlots(m2, smooth=T)
```



```
##          Test stat Pr(>|Test stat|)
## Wealth      0.6940      0.487850
## Pop        -1.1320      0.257936
## PInter       4.1351     3.878e-05 ***
## Density     -1.5723      0.116230
## Income       3.9914     7.100e-05 ***
## GrowthRate   2.8460      0.004527 **
## Tukey test    1.0944      0.273775
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

By analyzing the residual plots i see that the residuals for wealth, population, density are nearly zero but we are interested in the fitted (tukey test) of residuals where the p value is 0.46 which is not significant. so, there is no evidence to reject the null hypothesis. According to our assumption our fitted model has less residual values. Which helps us to predict the model for the years 2005 and 2025.

By using the table from the question i predict the fitted expenditure for the given years

FITTED VALUES CONFIDENCE INTERVAL (CONFIDENCE INTERVAL FOR A CONDITIONAL MEAN) -- Warwick 2005

```
predict(m2, newdata=data.frame(Pop=c(20442),
```

```

Wealth=c(85000),PInter=c(24.7),Density=c(214), Income=c(19500),
GrowthRate=c(35)), interval="confidence", level=0.95)

##          fit          lwr          upr
## 1 312.2412 279.7093 344.773

# SINGLE VALUE CONFIDENCE INTERVAL -- Warwick 2005
predict(m2, newdata=data.frame(Pop=c(20442),
Wealth=c(85000),PInter=c(24.7),Density=c(214), Income=c(19500),
GrowthRate=c(35)),interval="prediction", level=0.95)

##          fit          lwr          upr
## 1 312.2412 -24.28736 648.7697

# FITTED VALUES CONFIDENCE INTERVAL (CONFIDENCE INTERVAL FOR A CONDITIONAL
MEAN) -- Warwick 2025
predict(m2, newdata=data.frame(Pop=c(31033),
Wealth=c(89000),PInter=c(26),Density=c(325), Income=c(20000),
GrowthRate=c(40)), interval="confidence", level=0.95)

##          fit          lwr          upr
## 1 316.8208 280.1543 353.4873

# SINGLE VALUE CONFIDENCE INTERVAL -- Warwick 2025
predict(m2, newdata=data.frame(Pop=c(31033),
Wealth=c(89000),PInter=c(26),Density=c(325), Income=c(20000),
GrowthRate=c(40)), interval="prediction", level=0.95)

##          fit          lwr          upr
## 1 316.8208 -20.13254 653.7741

#####
# FITTED VALUES CONFIDENCE INTERVAL (CONFIDENCE INTERVAL FOR A CONDITIONAL
MEAN) -- Manroe 2005
predict(m2, newdata=data.frame(Pop=c(10496),
Wealth=c(58000),PInter=c(8.8),Density=c(695), Income=c(17100),
GrowthRate=c(35)), interval="confidence", level=0.95)

##          fit          lwr          upr
## 1 234.767 207.3638 262.1703

# SINGLE VALUE CONFIDENCE INTERVAL -- Manroe 2005
predict(m2, newdata=data.frame(Pop=c(10496),
Wealth=c(58000),PInter=c(8.8),Density=c(695), Income=c(17100),
GrowthRate=c(35)), interval="prediction", level=0.95)

##          fit          lwr          upr
## 1 234.767 -101.3045 570.8385

# FITTED VALUES CONFIDENCE INTERVAL (CONFIDENCE INTERVAL FOR A CONDITIONAL
MEAN) -- Manroe 2025
predict(m2, newdata=data.frame(Pop=c(13913),

```

```

Wealth=c(60000),PInter=c(10.1),Density=c(959), Income=c(18100),
GrowthRate=c(35)), interval="confidence", level=0.95)

##          fit          lwr          upr
## 1 233.8774 203.2047 264.5502

# SINGLE VALUE CONFIDENCE INTERVAL -- Manroe 2025
predict(m2, newdata=data.frame(Pop=c(13913),
Wealth=c(60000),PInter=c(10.1),Density=c(959), Income=c(18100),
GrowthRate=c(35)), interval="prediction", level=0.95)

##          fit          lwr          upr
## 1 233.8774 -102.4764 570.2313

#####

# FITTED VALUES CONFIDENCE INTERVAL (CONFIDENCE INTERVAL FOR A CONDITIONAL
MEAN) -- Tuxedo 2005
predict(m2, newdata=data.frame(Pop=c(10685),
Wealth=c(116000),PInter=c(6.1),Density=c(249), Income=c(28300),
GrowthRate=c(300)), interval="confidence", level=0.95)

##          fit          lwr          upr
## 1 -30.41357 -225.3316 164.5044

# SINGLE VALUE CONFIDENCE INTERVAL -- Tuxedo 2005
predict(m2, newdata=data.frame(Pop=c(10685),
Wealth=c(116000),PInter=c(6.1),Density=c(249), Income=c(28300),
GrowthRate=c(300)), interval="prediction", level=0.95)

##          fit          lwr          upr
## 1 -30.41357 -417.9521 357.125

# FITTED VALUES CONFIDENCE INTERVAL (CONFIDENCE INTERVAL FOR A CONDITIONAL
MEAN) -- Tuxedo 2025
predict(m2, newdata=data.frame(Pop=c(29246),
Wealth=c(115000),PInter=c(7),Density=c(656), Income=c(25000),
GrowthRate=c(100)), interval="confidence", level=0.95)

##          fit          lwr          upr
## 1 279.4956 209.0685 349.9227

# SINGLE VALUE CONFIDENCE INTERVAL -- Tuxedo 2025
predict(m2, newdata=data.frame(Pop=c(29246),
Wealth=c(115000),PInter=c(7),Density=c(656), Income=c(25000),
GrowthRate=c(100)), interval="prediction", level=0.95)

##          fit          lwr          upr
## 1 279.4956 -62.78072 621.772

Finale <- as.table(rbind(c(351.0252, 362.8376), c(274.014, 274.8319 ),
c(371.533, 409.1213)))

```

```
dimnames(Finale) <- list(Town = c("Warwick", "Manroe", "Tuxedo"),  
                          Expenditure = c("2005", "2025"))  
# According to my prediction for constructing the new houses by the  
municipality has to spend the expenditure per capita  
# amount in the specific years as follows:
```

Finale

```
##           Expenditure  
## Town           2005      2025  
##   Warwick 351.0252 362.8376  
##   Manroe  274.0140 274.8319  
##   Tuxedo  371.5330 409.1213
```